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create database Assignment_joins;
use Assignment_joins;

CREATE TABLE departments (
dept_id INT PRIMARY KEY,
dept_name VARCHAR(50),
location VARCHAR(50)
);
INSERT INTO departments VALUES
(10, 'HR', 'Jaipur'),
(20, 'IT', 'Bangalore'),
(30, 'Finance', 'Mumbai'),
(40, 'Marketing', 'Delhi'),
(50, 'Legal', 'Pune');
CREATE TABLE employees (
emp_id INT PRIMARY KEY,
emp_name VARCHAR(50),
dept_id INT,
salary INT,
city VARCHAR(50)
);
INSERT INTO employees VALUES
(101, 'Amit', 10, 45000, 'Jaipur'),
(102, 'Bhavna', 20, 72000, 'Bangalore'),
(103, 'Chirag', 20, 68000, 'Bangalore'),
(104, 'Divya', 30, 55000, 'Mumbai'),
(105, 'Esha', NULL, 39000, 'Delhi'),
(106, 'Farhan', 40, 61000, 'Delhi'),
(107, 'Gaurav', 60, 50000, 'Kolkata'),
(108, 'Hina', 30, 83000, 'Mumbai');
CREATE TABLE projects (
proj_id INT PRIMARY KEY,
proj_name VARCHAR(50),
dept_id INT,
budget INT
);
INSERT INTO projects VALUES
(1, 'Payroll Revamp', 10, 120000),
(2, 'Cloud Migration', 20, 500000),
(3, 'Mobile App', 20, 300000),
(4, 'Audit Automation', 30, 250000),
(5, 'Brand Refresh', 40, 180000),
(6, 'Data Lake', 70, 400000);

Select * from departments;
Select * from Employees;
Select * from projects;

-- q1
Select e.emp_id, e.emp_name, d.dept_name, d.location
from Employees as e
Join Departments as d
on e.dept_id = d.dept_id;

-- q2
Select p.proj_id, p.proj_name, d.dept_name, p.budget
from Departments as d
Join Projects as p
on d.dept_id = p.dept_id;

-- q3
Select e.emp_name, d.dept_name, p.proj_name

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from Departments as d
join Projects as p
join Employees as e
on d.dept_id = p.dept_id and p.dept_id = e.dept_id;
```

```
-- q4
Select e.emp_id, e.emp_name, e.dept_id, d.dept_name
from Employees as e
left join Departments as d
on e.dept_id = d.dept_id;
```

```
-- q5
Select d.dept_id, d.dept_name, p.proj_name, p.budget
from Departments as d
left join Projects as p
on d.dept_id = p.dept_id;
```

```
-- q6
Select e.emp_id, e.emp_name, e.dept_id
from Employees as e
left join Departments as d
on e.dept_id = d.dept_id
where d.dept_id is null;
```

```
-- q7
Select e.emp_name, d.dept_id, d.dept_name
from Employees as e
Right join Departments as d
on e.dept_id = d.dept_id;
```

```
-- q8
Select d.dept_name, p.proj_id, p.proj_name, p.dept_id
from Departments as d
right join Projects as p
on d.dept_id = p.dept_id;
```

```
-- q9
Select d.dept_id, d.dept_name, d.location
from Departments as d
Left join Employees as e
on d.dept_id = e.dept_id
where e.emp_id is null;
```

```
-- q10
Select *
from Employees as e
cross join Departments as d
where e.salary>70000 and (d.dept_name = 'Legal' or d.dept_name = 'Finance');
```

```
-- q11
select p.proj_name, p.budget, d.dept_name, d.location
from Projects as p
cross Join Departments as d
where d.location in ('Mumbai', 'Pune') and p.budget>=400000;
```

```
-- q12
Select count(*) as 'Total_combinations'
from Employees as e
```

```
cross join Departments as d;
```

```
-- q13
```

```
Select *  
from Employees as e  
where e.salary>60000;
```

```
-- q14
```

```
Select e.emp_id, e.emp_name, e.salary, e.city  
from Employees as e  
Where city in ('Mumbai', 'Delhi');
```

```
-- q15
```

```
Select emp_id, emp_name, dept_id, salary  
from Employees  
where dept_id is null;
```

```
-- q16
```

```
Select Dept_id, Dept_name, location  
from Departments  
where location = 'Bangalore' or location = 'Delhi';
```

```
-- q17
```

```
Select dept_id, dept_name, location  
from Departments  
where dept_id>=30;
```

```
-- q18
```

```
Select *  
from Departments  
where dept_name like 'F%'  
or Dept_name like 'L%';
```

```
-- q19
```

```
Select *  
from projects  
where budget between 200000 and 400000;
```

```
-- q20
```

```
Select *  
from projects  
where dept_id = 20;
```

```
-- q21
```

```
Select *  
from Projects  
where proj_name like '%a%' and budget<300000;
```

```
-- q22
```

```
Select e.emp_name, d.dept_name, d.location, p.proj_name, p.budget  
from Employees as e  
join Departments as d  
join Projects as p  
on e.dept_id = d.dept_id and d.dept_id = p.dept_id;
```

```
-- q23
```

```
Select e.emp_id, e.emp_name, d.dept_name, p.proj_name  
from Employees as e  
left join Departments as d
```

```
        on e.dept_id = d.dept_id
left join Projects as p
        on d.dept_id = p.dept_id;
```

-- q24

```
Select e.emp_name, e.salary, d.dept_name, p.proj_name, p.budget
from Employees as e
left join Departments as d
        on e.dept_id = d.dept_id
left join Projects as p
        on d.dept_id = p.dept_id
where e.salary>60000 and p.budget>250000;
```

-- q25

```
Select d.dept_id, d.dept_name, e.emp_name, p.proj_name
from Departments as d
left join Employees as e
        on e.dept_id = d.dept_id
left join Projects as p
        on d.dept_id = p.dept_id;
```

-- q26

```
Select e.emp_name, d.location, p.proj_name
from Employees as e
left join Departments as d
        on e.dept_id = d.dept_id
left join projects as p
        on d.dept_id = p.dept_id
where d.location in ('Bangalore', 'Mumbai');
```

-- q27

```
Select distinct(e1.emp_name), e2.emp_name, e1.city
from Employees as e1
join Employees as e2
        on e1.city = e2.city
where e1.emp_id < e2.emp_id;
```

-- q28

```
Select p.proj_name, p.dept_id, e.emp_name
from Projects as p
join Employees as e
        on p.dept_id = e.dept_id
left join Departments as d
        on e.dept_id = d.dept_id;
```